

Department of Computer Science & Engineering

Data Mining Techniques

Module-2 bank

1. a) Consider the training examples in the following table for a binary classification problem.

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

- Compute the Info gain for the overall collection of training examples.
 - Compute the Info gain for the Customer ID attribute.
 - Compute the Infogain for the Gender attribute.
 - Which attribute is selected as root of the tree?
- b) How do different pruning strategies impact the computational complexity of training a decision tree?

2) Consider the following dataset of popularity of playlists on a streaming platform:

	Number of Tracks	Genres Covered	Collaborative	Average Track Popularity	Playlist Duration	Featured Artist	Time of Creation	Day of Creation
0	22	5	No	86	128	Yes	Morning	Weekday
1	41	5	Yes	70	49	No	Afternoon	Weekday
2	29	6	No	32	270	Yes	Evening	Weekend
3	5	7	No	35	90	No	Morning	Weekend
4	19	8	Yes	34	207	Yes	Night	Weekend
5	13	8	Yes	34	91	Yes	Evening	Weekday
6	19	2	No	83	174	Yes	Evening	Weekday
7	28	9	Yes	76	245	Yes	Night	Weekend
8	8	7	Yes	78	32	No	Morning	Weekday
9	16	4	Yes	38	189	No	Morning	Weekday

- How does the choice of splitting criteria (e.g., Gini impurity vs. entropy) affect the performance of the decision tree in predicting playlist popularity? Which criterion yields the best results for this dataset?
- How are continuous variables like 'Number of Tracks' and 'Average Track Popularity' handled in the decision tree induction process? What impact does the method of handling these variables have on the model's performance?
- What is the optimal depth of the decision tree for this prediction task? How does changing the depth affect the model's accuracy, and what signs indicate overfitting or underfitting at certain depths?
- How does the decision tree model handle categorical variables like 'Time of Creation' and 'Day of Creation'? Are certain split strategies (binary vs. multi-way splits) more effective for these types of variables?
- Would discretizing continuous features into bins improve the decision tree's performance or interpretability? If so, how should the bins be defined?

3) Based on the provided training data of various animals with features and their corresponding classes (mammals or non-mammals) answer the following questions

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

- Using the Naïve Bayes classifier, classify a new animal that gives birth, does not fly, lives in water, and does not have legs. Compute the probability that it is a mammal or non-mammal.
- Given that an animal can fly, find the posterior probability that it belongs to the mammal class.
- If an animal lives in water and does not have legs, find the probability that it gives birth instead of laying eggs.

4) Consider the following dataset

Age Group	Symptoms	Diagnosis	Treatment Success
Child	Fever	Viral	Yes
Adult	Cough	Bacterial	No
Senior	Fatigue	Other	Yes
Adult	Fever	Viral	Yes
Child	Cough	Viral	No
Senior	Fever	Bacterial	Yes
Adult	Fatigue	Other	No
Child	Fever	Bacterial	Yes
Senior	Cough	Viral	No
Adult	Fever	Other	Yes
Child	Fatigue	Bacterial	No
Senior	Fever	Viral	Yes
Adult	Cough	Viral	Yes
Child	Fever	Other	No

- i. What is the probability of treatment success for a child with a fever and a viral diagnosis?
- ii. How likely is it that an adult with cough symptoms has a bacterial diagnosis given that their treatment was not successful?
- iii. For a senior presenting with fatigue, what is the likelihood of the diagnosis being 'Other' if we know their treatment was successful?
- iv. Given a patient has been diagnosed with a bacterial infection, what is the probability of treatment success for each age group (Child, Adult, Senior)?
- v. If a patient's treatment was unsuccessful, what is the probability distribution across different diagnoses (Viral, Bacterial, Other)?
- vi. For a patient with fever symptoms, how does the probability of treatment success vary by diagnosis (Viral, Bacterial, Other)?
- vii. What is the overall probability of treatment success for patients with cough symptoms, regardless of age group or diagnosis?
- viii. Considering all patients with a 'Viral' diagnosis, what is the probability distribution of symptoms (Fever, Cough, Fatigue)?

5) Consider a dataset with two features (X_1 , X_2) and a binary target variable. The dataset consists of four data points:

A (1, 2) with class +1

B (3, 3) with class +1

C (2, 4) with class -1

D (5, 2) with class -1

E (4, 4) with class +1

F (6, 5) with class -1

G (2, 1) with class +1

H (5, 4) with class -1

- a) Manually plot these points on a 2D graph, marking the +1 class with one color or marker and the -1 class with another.
- b) Based on the plot, identify which points could potentially be the support vectors for the SVM decision boundary.
- c) Knowing that the decision boundary for a linear SVM in a 2D space can be represented as $x_1 + w_2 x_2 + b = 0$, use the potential support vectors to formulate a system of equations.
- d) Solve the system of equations to find the values of w_1 , w_2 , and b . This requires setting up equations based on the support vectors lying on the margin lines, where the decision function is equal to 1 for the positive class and -1 for the negative class.

6) a) Consider the one-dimensional data set shown in the following Table.

Table 5.4. Data set for Exercise 13.

x	0.5	3.0	4.5	4.6	4.9	5.2	5.3	5.5	7.0	9.5
y	−	−	+	+	+	−	−	+	−	−

- Classify the data point $x = 5.0$ according to its 1-, 3-, 5-, and 9-nearest neighbours (using majority vote).
- Repeat the previous analysis using the distance-weighted voting approach
- The NN classifier doesn't build a model, so it should be fast. However, on the contrary, it is slow learner why?

b) Given row entry, count represents the number of data tuples having the values for department, status, age, and salary given in that row.

<i>department</i>	<i>status</i>	<i>age</i>	<i>salary</i>	<i>count</i>
sales	senior	31...35	46K...50K	30
sales	junior	26...30	26K...30K	40
sales	junior	31...35	31K...35K	40
systems	junior	21...25	46K...50K	20
systems	senior	31...35	66K...70K	5
systems	junior	26...30	46K...50K	3
systems	senior	41...45	66K...70K	3
marketing	senior	36...40	46K...50K	10
marketing	junior	31...35	41K...45K	4
secretary	senior	46...50	36K...40K	4
secretary	junior	26...30	26K...30K	6

Let status be the class label attribute.

- Design a multilayer feed-forward neural network for the given data. Label the nodes in the input and output layers.
- Using the multilayer feed-forward neural network obtained above, show the weight values after one iteration of the backpropagation algorithm, given the training instance "(sales, senior, 31...35, 46K...50K)". Indicate your initial weight values and biases, and the learning rate used.

7)

Hypothetical Similarity Matrix (Table 3):

	Point A	Point B	Point C	Point D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

In this matrix, the values represent the distance between points (e.g., the distance between Point A and Point B is 2). A lower value indicates higher similarity (or smaller distance).

- Apply the single-link hierarchical clustering method using the similarity matrix. Which two points would you first merge based on this approach?
- Apply the complete-link hierarchical clustering method to the same matrix. Which two points would you first merge based on this approach?
- Create a dendrogram that represents the clustering process you performed with the complete-link method and single linkage, showing the order of merges and the distances.
- Compare and contrast the dendrograms produced by the single-link and complete-link methods. How do the methods differ in terms of the clusters they form with this data?

8) a) Consider the following one-dimensional data set: 12, 22, 2, 3, 33, 27, 5, 16, 6, 31, 20, 37, 8 and 18. Given $k=2$ and initial cluster centres to be 5, 6 and 31. What are the final cluster centres obtained on applying the k-means algorithm?

b) Use the DBSCAN algorithm to cluster the following points $\{a[2,3], b[2,4], c[3,4], d[5,6], e[6,7], f[5,7], g[6,3]\}$, assume Manhattan distance as a distance measure, ϵ as 2 and MinPts as 3. Suppose if there are 100 points in a 2-D area. All points are uniformly spaced. What is the expected result from running DBSCAN on this data.

9)

Supermarket Customer Dataset:

Customer ID	Total Spend	Visit Frequency	Basket Variety	Average Basket Size	Customer Type
1	520	5	12	6	Regular
2	300	2	8	10	Occasional
3	650	10	15	8	Regular
4	150	1	5	5	First-time
5	450	6	10	9	Regular
6	200	3	7	4	Occasional
7	700	12	18	12	Regular
8	100	1	3	3	First-time
9	400	4	9	7	Regular
10	250	2	6	5	Occasional

- a) How will you determine the optimal number of clusters (k) for the K-Medoids algorithm given this supermarket dataset?
- b) Which distance metric would you choose to calculate the dissimilarity between customers, and why is it appropriate for this dataset?
- c) Since Customer Type is a categorical feature, what encoding method will you use to make it compatible with the K-Medoids algorithm?
- d) Describe the steps to perform K-Medoids clustering on this dataset. How does the algorithm select initial medoids and iterate to improve the clustering?
- e) After applying K-Medoids, how would you interpret the resulting clusters in terms of customer shopping behavior?

10)

The table below is a distance matrix for 6 objects.

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single link by drawing a dendrogram.
- Show the final result of hierarchical clustering with complete link by drawing a dendrogram
- Change two values from the matrix so that your answer to the last two question would be same.

b) Which clustering method should we use?

Which clustering method(s) is most likely to produce the following results at $k = 2$? Choose the most likely method(s) and briefly explain why it/they will work better where others will not in at most 3 sentences.

Here are the five clustering methods you can choose from:

- Hierarchical clustering with single link
- Hierarchical clustering with complete link
- Hierarchical clustering with average link
- K-means
- GMM (with no assumption on the covariance matrices)

